

TULGA, Turkey

WMO: 171970

Lat: 38.350N

Lon: 38.250E

Elev: 3054

StdP: 13.14

Time Zone: 3.00 (E03)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	22.8	6.9	8.8	28.3	10.4	10.5	30.3	17.6	37.1	14.4	36.3	4.7	180	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.0	99.6	63.7	97.1	63.6	94.9	63.5	68.8	90.6	67.1	89.9	65.9	89.3	6.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	90.4	78.7	58.9	83.1	77.1	57.0	77.6	75.4	34.9	90.8	33.4	90.2	32.3	89.4	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.2	10.8	14.8	103.6	4.5	2.1	11.6	105.1	9.0	106.4	6.4	107.6	3.2	109.1		
(5)			13.4	71.1	4.1	3.5	10.5	73.6	8.1	75.6	5.7	77.5	2.8	80.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	34.7	39.7	47.6	56.6	65.5	75.7	81.9	82.8	73.2	61.0	47.4	38.0		
	DBStd	17.53	6.36	6.67	6.41	5.95	5.80	4.87	4.18	4.13	5.98	5.61	5.63	6.18		
	HDD50	1390	475	292	121	14	0	0	0	0	0	1	113	374		
	HDD65	4037	940	710	538	258	66	0	0	0	0	148	528	838		
	CDD50	4591	0	3	48	211	482	770	988	1016	697	341	34	1		
	CDD65	1763	0	0	0	5	83	320	523	551	258	23	0	0		
	CDH74	21126	0	0	0	56	800	3519	6703	7150	2691	207	0	0		
	CDH80	10688	0	0	0	2	211	1572	3717	4019	1145	22	0	0		
	WSAvg	4.2	3.3	3.5	4.9	4.8	4.8	5.3	5.2	4.8	4.5	3.5	2.9	3.0		
	PrecAvg	14.3	1.7	1.8	1.6	1.9	1.7	0.5	0.1	0.1	0.4	1.4	1.6	1.6		
(15) Precipitation	PrecMax	19.6	3.6	3.4	4.3	3.8	6.2	2.3	0.5	0.4	1.5	3.4	3.7	3.9		
	PrecMin	9.6	0.2	0.4	0.1	0.4	0.2	0.0	0.0	0.0	0.0	0.3	0.0	0.0		
	PrecStd	2.3	0.8	0.8	1.1	0.9	1.1	0.5	0.1	0.1	0.4	0.9	1.0	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	62.4	69.4	79.0	89.3	97.5	102.5	102.6	96.5	82.6	67.7	57.4		
		MCWB	43.5	48.7	52.6	55.5	61.1	64.1	63.4	64.4	60.6	59.4	54.0	46.8		
	2%	DB	48.3	56.8	65.8	75.3	85.4	94.2	100.0	100.2	93.3	79.1	63.5	51.5		
		MCWB	41.3	46.6	50.6	54.5	59.7	63.8	63.1	64.4	61.3	58.2	50.2	44.1		
	5%	DB	45.5	53.2	62.6	71.9	82.2	91.4	97.1	97.9	89.8	76.1	60.6	48.5		
		MCWB	39.7	44.4	48.9	53.7	59.4	62.7	63.4	64.1	61.5	57.9	48.9	42.8		
	10%	DB	43.1	49.8	59.2	68.8	78.7	88.3	94.7	95.3	86.6	72.9	57.2	46.3		
		MCWB	38.3	42.3	47.5	52.7	58.4	62.1	63.4	63.8	60.9	56.4	48.2	42.0		
	0.4%	WB	44.9	49.7	55.4	60.8	64.8	69.1	70.9	73.0	67.7	63.3	56.4	48.9		
		MCDB	48.7	60.5	65.0	72.3	82.2	89.1	90.5	89.5	86.2	75.0	64.2	55.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.6	47.4	52.2	57.5	62.6	66.6	68.3	69.3	65.6	60.9	52.8	46.3		
		MCDB	46.9	55.1	62.3	69.5	80.2	88.6	90.2	91.8	85.5	74.1	58.1	49.3		
	5%	WB	40.7	45.3	50.1	55.6	60.8	65.2	66.7	67.5	63.9	59.2	50.8	44.2		
		MCDB	44.4	51.9	60.3	67.5	77.0	86.6	89.5	91.4	84.3	72.1	57.2	47.5		
	10%	WB	39.0	43.2	48.1	53.9	59.3	63.9	65.4	66.0	62.1	57.7	49.0	42.5		
		MCDB	42.3	48.6	58.1	65.5	74.8	84.4	89.0	91.3	82.7	69.3	55.9	45.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	14.3	17.0	19.6	21.7	23.9	24.7	25.0	24.3	19.7	17.1	12.0		
		MCDBR	14.3	19.4	21.6	24.3	25.4	25.8	26.9	26.9	27.6	24.4	21.8	15.4		
		MCWBR	9.3	11.6	11.4	11.2	10.3	9.3	9.4	9.5	10.7	11.3	12.1	10.0		
		MCDBR	12.0	17.2	19.5	20.4	23.1	24.2	23.7	24.6	24.5	20.5	17.5	11.4		
	5% WB	MCWBR	8.6	10.9	11.2	10.7	10.3	10.3	10.3	11.2	11.7	10.7	10.9	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.365	0.407	0.457	0.427	0.372	0.362	0.378	0.348	0.390	0.345	0.327		
	taud		2.244	2.154	2.083	1.984	2.134	2.326	2.360	2.295	2.368	2.197	2.294	2.302		
	Ebn at Noon		263	268	268	261	272	287	289	281	285	256	256	256		
	Edh at Noon		32	40	47	56	49	40	39	40	35	38	30	28		
(44) All-Sky Solar Radiation	RadAvg		674	981	1347	1689	2071	2502	2546	2254	1845	1239	845	592		
	RadStd		91	161	125	158	128	114	69	74	75	95	99	122		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	-1.58	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page